

Derek Gallagher

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EDUCATION

Concordia University

Montreal, QC

Bachelor of Engineering in Software Engineering (Co-op)

Expected 2027

- **Relevant Coursework:** Object-Oriented Programming, Data Structures and Algorithms, Web-based Enterprise Application Design
- **GPA:** 3.6

EXPERIENCE

Business Application Development Intern

Jan 2026 – Apr 2026

National Research Council Canada

Ottawa, ON

- Supported **CMDB data integration** through **iPaaS imports** of legacy and new configuration items.
- Designed **SQL views** and **Power BI dashboards** to track ticket KPIs, reducing manual reporting.
- Resolved IT service tickets involving access management, permissions, and updates across enterprise systems.

SAP Functional Intern

May 2025 – Aug 2025

National Research Council Canada

Ottawa, ON

- Designed and deployed **Power BI dashboards** to monitor **20000+ service tickets**, reducing manual reporting time by **60%**.
- Assisted in **SAP FSA and DAA module testing** for travel operations on **SAP ECC**.
- Collaborated with the **Service Management Office** to implement a new ticketing workflow, applying change management principles to improve response time by **25%**.

Software Quality Assurance Intern

Sep 2024 – Dec 2024

Matrox Video

Montreal, QC

- Executed and maintained **300+ test cases** in **Jira** and **TestRail**, identifying and reporting many defects.
- Ran validation tests on Matrox video boards across **3 operating systems** and **many kernel versions**.
- Developed **PowerShell scripts** to automate configuration tasks across hardware boards.
- Assembled 5 custom desktop systems, including **BIOS configuration** and **operating system installation**.

PROJECTS

AI-Powered CV Generator | *LLMs, NLP, Ollama, Java*

2025

- Designing an **AI-driven application** to generate tailored CVs from job postings and structured data.
- Implementing rule-based logic to align skills, projects, and keywords with role requirements.
- Exploring **local LLM integration** for fast, privacy-preserving CV customization workflows.

Hospital Data Management System | *MySQL, MongoDB, Python*

2025

- Designed a normalized **MySQL** relational database from an ER model to store large-scale hospital data.
- Built an **ETL pipeline** in **Python** to migrate data from **MySQL** to a **MongoDB** document model, applying denormalization for scalable NoSQL queries.

Discord-like Communication Platform | *React, Python (Flask), MySQL, AWS, Bootstrap*

2025

- Developed a real-time messaging app supporting **100+ concurrent users** with text, voice, and media sharing.
- Implemented **WebSocket-based chat channels**, achieving message latency under **200ms**.
- Deployed application to **AWS** and optimized database queries for faster load times.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, Python, C, C#, C++, PHP, R, SQL (MySQL, PostgreSQL)

Web & Frameworks: React (Vite), Node.js, Express.js, Flask, Java Servlets, JSP, Bootstrap, jQuery, Three.js

Operating Systems: Windows 10/11, Linux (Ubuntu)

Databases & Data: MySQL, PostgreSQL, MongoDB, Prisma (ORM), Relational Schema Design, JSON, AWS

Tools & Platforms: GitHub, Power BI, Jira, TestRail, SAP ECC, Blender, Ollama, Langchain

Spoken Languages: English (Fluent), French (Fluent)